LOGICAL POSITIVISM

The marriage of empiricism with mathematical logic was completed in Vienna around 1930 by a group of philosophers who had developed a viewpoint called logical positivism. A major tenet of positivism is that only directly observable objects and events should be considered valid scientific subject matter, making it a highly empiricist philosophy. Anything that can't be observed is, for the positivist, unwanted metaphysical baggage. The task of science, in this view, is to ascertain the logical relationships between all of these observables. The network of logical connections built up in this way is a scientific theory, and such a theory is the only "positive" knowledge that we can possibly have. The clarity and precision of positivism were attractive to many philosophers. Nothing is more clear and unambiguous than mathematical logic; and empirical facts are certainly a firm foundation upon which to build science. But there is a problem: many things in the sciences are not directly observable. What do we do about these things? (The grandfather of positivism, Ernst Mach, had stoutly denied the existence of atoms because they could not be observed.) To address this problem, the logical positivists demanded "rules of correspondence" between theoretical entities and empirical observations. As long as a concept could be connected to observations by a set of rigid rules, the concept was legitimate. If such rules can't be found, get rid of the concept.

A similar kind of thinking is found in the movement known as operationalism. The key idea here is the "operational definition" of a scientific concept, which is a definition purely in terms of operations resulting in a measurement. For example, Newton had defined time with a verbal statement that he regarded as self-evident, but that was operationally meaningless. Einstein's redefinition of time in terms of the operations with clocks that are needed to measure time turned out to be an important advance. Using examples like this, P. W. Bridgman (an experimental physicist with philosophical interests) formulated the idea of an operational definition. Bridgman claimed that any concepts that couldn't be defined operationally should be considered scientifically meaningless. The spirit of operationalism is quite consistent with logical positivism, and both were heavily influenced by Mach. In terms of the positivist program, operational definitions are a particular kind of correspondence rule. Logical positivism dominated the philosophy of science for decades, and still exerts some influence. But it came under attack around the middle of the century, and has waned in authority since then. We'll see some of the problems with positivism as we go on with the chapter. A different philosophical vision of science, however, had been articulated even before positivism was invented. The brilliant French physicist and philosopher, Henri Poincaré, proposed that science needs to employ "conventions" that are creations of the human mind, not directly tied to empirical facts. These conventions are not merely free creations, however, because they are constrained by our observations of the world in a variety of indirect ways, and sometimes must change with the course of scientific progress. Poincaré, writing near the turn of the century, devised a philosophy of science that in many respects foreshadowed much later developments.

FAILURE OF INDUCTION

Logical positivism has a problem, even if you accept the pristine theory-independence of facts: the problem of induction. On the one hand, there is a purely logical problem with inductivism. Any logical proof of the inductivist premise must be itself a proof by induction ("induction has always worked, therefore it is valid"). This obviously is no proof at all, since it assumes the truth of the premise it sets out to prove. On the other hand, there is a more practical problem: the conclusions based on induction do, in fact, sometimes turn out to be wrong. For example, Lavoisier had shown (by the late eighteenth century) that a number of substances are elements, that is, substances that cannot be further broken down or changed to another substance. By the beginning of the twentieth century, this property of elements (unchangeability) had been shown empirically to be true innumerable times. That elements are immutable was taken to be a well-known fact; note that the conclusion is clearly based on induction.

Around 1900, radioactivity was discovered, and Rutherford demonstrated that the radioactive decay process is actually a change from one element to another. The previous conclusion, based on induction, had simply been wrong.

FALSIFICATIONISM

To avoid the problem of induction, we can turn the question around: Once a theory has been proposed, we demand that it make statements that can be falsified by comparison with observation. We then make the observations. If they don't agree, that is, if the statement is false, we pronounce the theory wrong. If the observation agrees with the prediction (statement is true), then we say the theory is confirmed by the observation. But we don't necessarily say that the theory is true. We've made a subtle but important change in our philosophical view concerning the goals of science. In this picture, we accept theories on a provisional basis, as long as they continue to be confirmed. We can never prove a theory is right, only that it's wrong (if a statement is false). This idea has been implicit in science for some time, but as a formal doctrine it is closely associated with the philosophy of Karl Popper.

The criterion of falsifiability serves a number of worthwhile purposes. Primarily, it allows us, when deciding questions of validity, to weed out statements that have no observable consequences. An extreme example is the claim of an Aristotelian opponent of Galileo that he made after he (the opponent) observed through a telescope mountains and craters on the moon. To make this observation consistent with his belief that the moon is a perfect sphere, he hypothesized the existence of an invisible substance that covered the mountains and filled the craters, making a perfect spherical surface. This substance could not in principle be observed, so his claim could not be falsified. Such statements are of little use in the discourse of science.

But the rigorous logic of falsificationism is rarely enforced in practice; if it was, many productive and valid ideas would have died an early death. A famous example is the so-called stellar parallax predicted by the Copernican theory. Stellar parallax simply means that the stars should appear to be in slightly different places as the earth itself moves around from place to place. This sensible conclusion is hard to escape, but the effect was not observed until about 1840, roughly 300 years after the work of Copernicus. Strictly speaking, the theory was falsified. But it wasn't given up, nor should it have been. (The reason parallax was not seen is that the stars are much farther away than anyone suspected, and it awaited the technical development of better telescopes to make the observation.) The point here is that a good theory, which has a number of empirical confirmations and is leading progressively to new ideas and applications, will not and should not be given up even if a few minor discrepancies remain. Falsificationism is still a valuable methodological tool. The confirmation of bold new ideas, when put to the test of falsification, drives the progress of science (as in the case of bending starlight predicted by Einstein's general relativity in 1915 and observed during a solar eclipse in 1919). Equally important, the falsification of well-accepted ideas, as in the radioactive transmutation of elements cited above, also drives the progress of science. So, falsificationism serves well as a framework for thinking about scientific questions, but it doesn't provide us with a universal principle for determining validity in science.

Gregory N Derry. What science is and how it works. Princeton University Press, 1999.